MINI GROUP PROJECT

PRINCIPLES OF DATA PRESENTATION

ANALYSIS OF PLAYER AND TEAM PERFORMANCE IN ICC WORLD CUP 2023

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Git hub Repository: **https://github.com/Ruzen-Maharjan/DS-ICC-World-Cup-2023-analysis.git**

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Abstract

This project analyzes the ICC ODI World Cup 2023 dataset to understand player and team performance. The dataset includes important variables such as runs scored, wickets taken, strike rate, batting average, and match results. The main aim of this analysis is to identify patterns in performance and understand what factors contribute to success in cricket matches.

From the analysis, it is observed that consistent performance is more important than aggressive short-term performance. Players like Virat Kohli showed high consistency, which helped them score more runs overall. Similarly, bowlers who maintained a low economy rate and applied continuous pressure played a key role in controlling matches.

Team performance analysis also showed that balanced teams with both strong batting and bowling tend to win more matches. For example, India had the highest win ratio, showing consistent overall performance.

Overall, this project helps in understanding how data analysis can be used in real-world sports to identify trends and improve decision-making.

Introduction & Background: (Vraj Gajera)

Cricket is one of the most popular sports worldwide, and the ICC One Day International (ODI) World Cup is held in different countries and is followed by millions of fans. Each match creates huge amounts of data, like player statistics, team performance, and final results. Because of this, it attracts analysts to apply their knowledge to analyze the data.

We choose the ICC ODI World Cup 2023 dataset because it provides a large amount of well-structured data that is helpful for analyzing the dataset. Working with this dataset helps us understand how data is used in real-world projects. There are many important variables used like wicket taker, runs scored, final result, and more, which helps us to analyze different part of the game.

From a data science point of view, this dataset is useful for performing data analysis, identifying key trends, and creating visualizations. These are commonly used by professionals to extract meaningful insights from data.

By analyzing this dataset, the main goal of this project is to evaluate both team and player performance and understand patterns that may influence match outcomes. In addition, this project helps us apply our knowledge and skills to a real-world sports dataset.

Methods

Data Collection:

The dataset used in this project is the ICC ODI World Cup 2023 dataset, which includes match-level and ball-by-ball data. The data was loaded using Python and analyzed using pandas.

Data Preparation:

Missing values in extra run columns such as wides, no-balls, byes, and leg-byes were replaced with zero. Wicket-related missing values were filled with 'None' to ensure consistency.

Variable Creation:

New variables such as total runs and economy rate were calculated. Batting statistics such as total runs, batting average, and strike rate were computed for each player.

Analytical Methods:

Descriptive statistics and visualizations were used to analyze the data. Bar charts were used to compare runs and wickets, while scatter plots were used to examine the relationship between batting average and strike rate. These methods help in identifying patterns and trends in player and team performance.

Analysis & Results: Batsman performance: (Kugathas)

While analyzing the performance of the top 10 highest run scorers, we found that Virat Kohli scored the highest number of runs, creating a large gap compared to other players. This indicates both strong performance and consistency throughout the tournament.

On the other hand, Rohit Sharma and Quinton de Kock also performed well, with their final run scores being very close to each other. Another observation is that many middle-ranked players scored a similar number of runs, meaning that even a small improvement in performance could significantly change their rankings. A key finding is that the highest run scorers were not necessarily the most aggressive players. Instead, they performed consistently in every match. This highlights that in modern cricket, consistent performance is more valuable than short-term aggressive play. (See Figure 1).

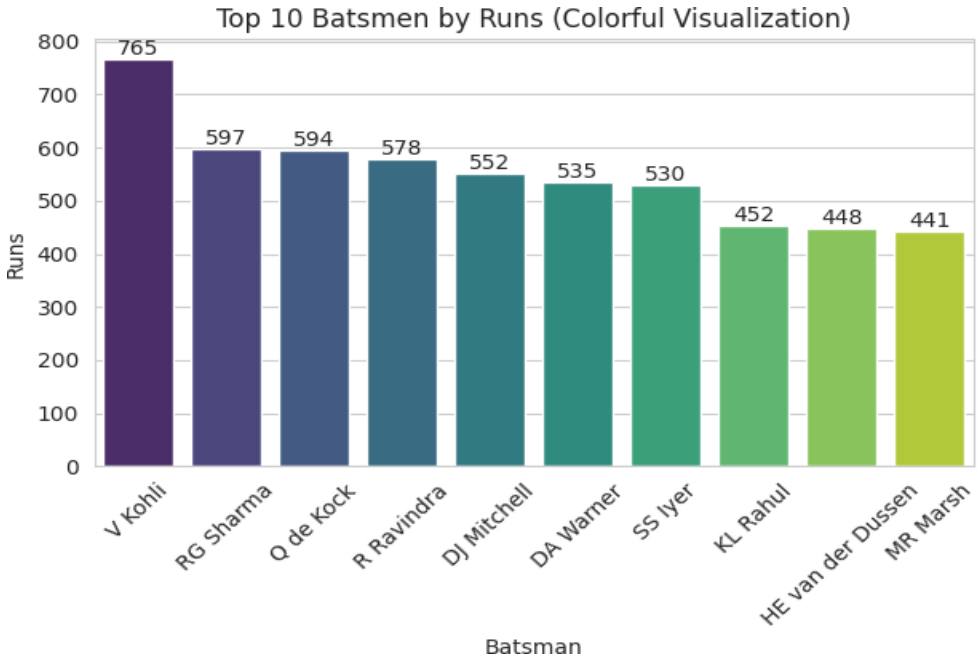

Figure 1: Top 10 Run Scorers in ICC World Cup 2023

Batting Average vs Strike Rate: (Kugathas)

When we compare Batting average and Strike Rate, we can see some most important output like players with higher strike rates did not score the highest runs because aggressive play increases the chances of getting out. On the other hand, players with higher averages played more carefully and scored more runs. For example, Virat Kohli shows that maintaining high average without taking risk, while Heinrich Klaasen gave more importance to fast scoring with less consistency. This shows that difference between risk and stability. Another important point is that those who performed well in one match are not as important as those who performed in every match and this helps the team in the long run.

(See Figure 2)

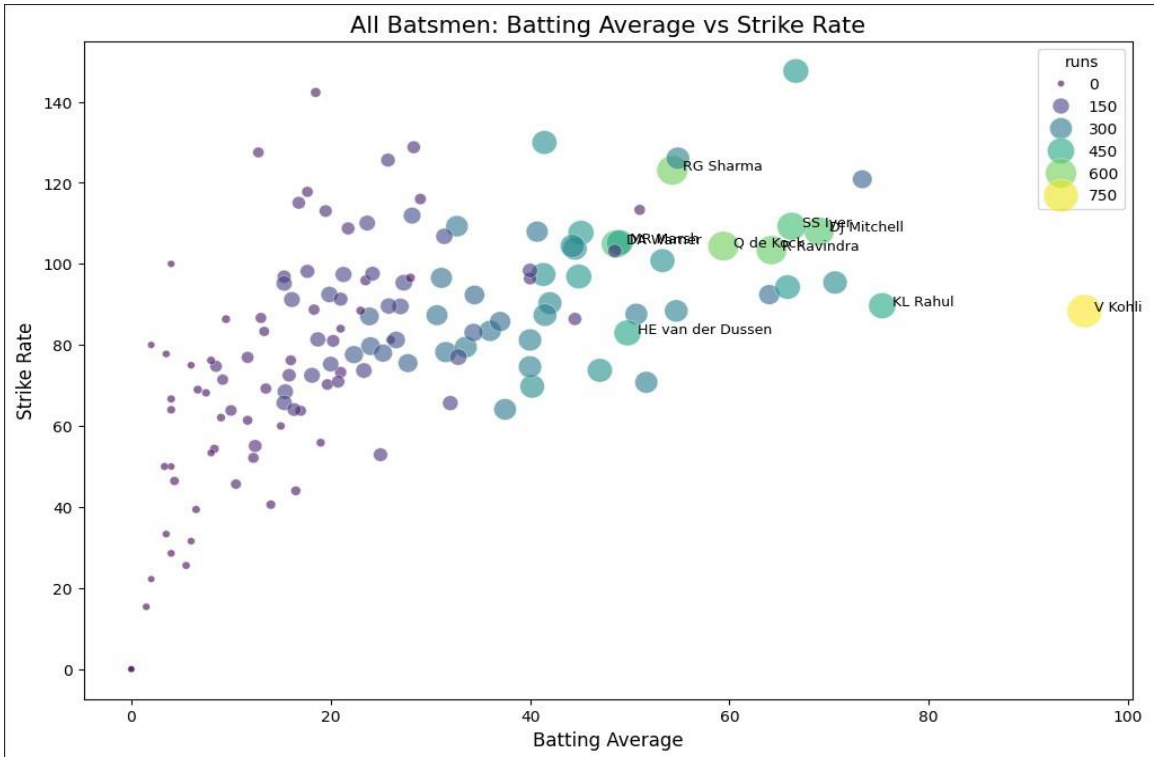

Figure 2: Batting Average vs Strike Rate of Players

Bowling performance by bowler in ICC ODI world cup: (Vraj Gajera)

When we closely see the graph given below and the dataset, it shows a very interesting pattern how wickets were taken by different players. Some bowlers played only a few matches; some bowlers took fewer wickets but still had very impactful performances. This clearly shows that bowlers have an ability to change match any time. A good example is Mohammed Shami.

(See Figure 3)

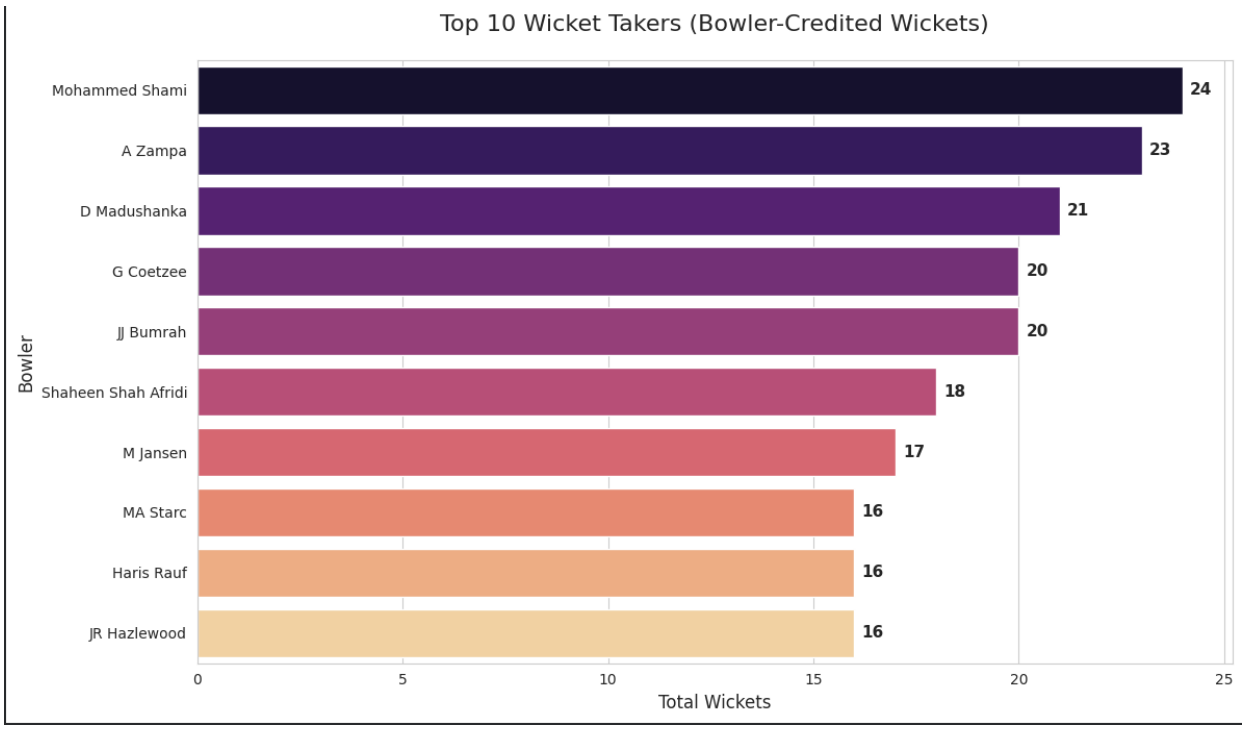

Figure 3: Bowling Performance (Wickets Taken by Bowlers)

Top 10 Bowlers by Economy Rate: (Ruzen)

From the chart, we can see that most of the top bowlers have similar economy rates, meaning the differences between them are small but still important. Jasprit Bumrah maintained a very low economy rate, which shows strong control over runs. It is important for a bowler to maintain a low economy rate consistently across overs rather than performing well in just one over. If a bowler loses control in a single over, it can negatively affect the team's overall performance and final score.

Another key point is that consistently applying pressure can force batsmen to take risks, which may lead to wickets.

(See Figure 4)

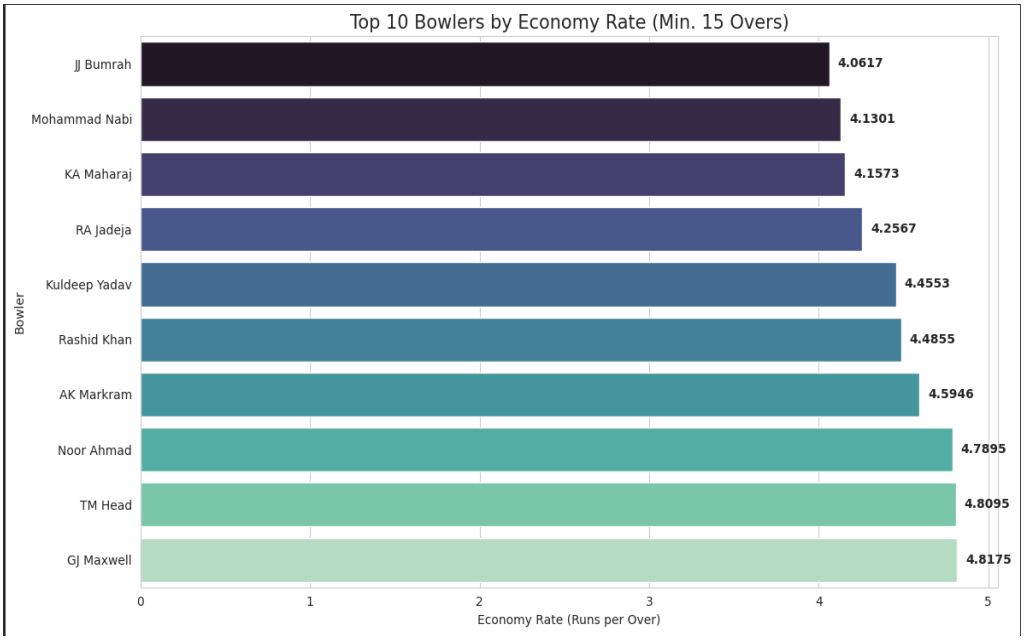

Figure 4: Top 10 Bowlers by Economy Rate

Key Things from Both Graphs: (Vraj Gajera)

Metric	Economy	Wicket
Bowling style	Spin (90%)	Pace (90%)
Common Player	JJ Bumrah	JJ Bumrah
Role	Pressure	Aggression

Team Performance Analysis: (Minhaj)

When we look at the column graph of win ratio, it is clear that India won the highest number of matches which also shows their good performance throughout the tournament. An important point is that teams that perform well and remain stable in both bowling and batting achieve better results. As seen from the above graphs, teams with consistent performance get better outcomes compared to those with inconsistent performance. Hence, those teams that maintain good balance and consistency are more likely to achieve success in the tournament.

In addition to this, the toss decision also plays a role in match outcomes. The distribution of toss decisions shows whether teams prefer to bat or bowl first. From the visualization, it can be observed that teams often choose to bowl first, which may be influenced by pitch conditions or match strategy. However, winning the toss does not always guarantee winning the match, which shows that overall team performance is still more important.

(See Figure 5 & 6)

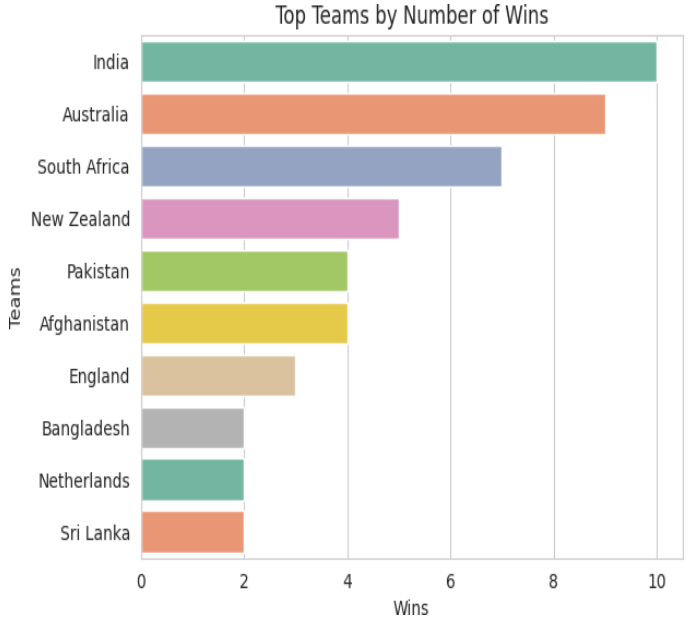

Figure 5: Team Win Ratio in ICC ODI World Cup 2023

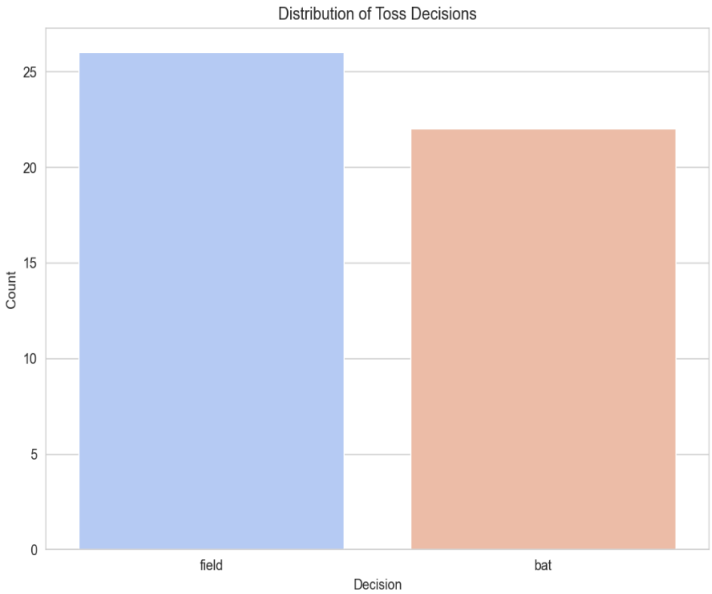

Figure 6: Distribution of Toss Decisions

Discussion

In this project, we analyzed different aspects of the ICC ODI World Cup 2023 dataset, including batting performance, bowling performance, and overall team results. The main aim was to understand what factors affect match outcomes and player success.

One of the main findings from the batting analysis is that consistency is more important than aggressive play. Players with high averages, like Virat Kohli, performed better overall compared to players who focused only on high strike rates. This shows that scoring steadily in every match is more valuable than scoring quickly in only a few matches.

From the bowling analysis, it is clear that both wicket-taking ability and economy rate are important. Some bowlers, such as Mohammed Shami, showed that taking multiple wickets in fewer matches can still have a strong impact. At the same time, bowlers like Jasprit Bumrah maintained low economy rates, which helped control the game by reducing runs. This balance between aggression and control is very important in bowling.

The team performance analysis showed that teams with balanced performance in both batting and bowling are more successful. India, for example, had the highest win ratio, which reflects their consistency throughout the tournament. This supports the idea that teamwork and stability are key factors in winning matches.

However, there are some limitations in this analysis. The dataset only includes one tournament, so the results may not apply to all cricket matches. Also, external factors like pitch conditions, weather, and player injuries were not considered, which can also affect performance.

CONCLUSION

In conclusion, this project shows that data analysis can provide useful insights into cricket performance. The results highlight that consistency in batting, control in bowling, and overall team balance are the key factors for success. Teams that maintain stable performance across matches are more likely to achieve better outcomes.

Moreover, this analysis is limited to one tournament and does not include external factors such as pitch conditions or player fitness. In future, using more datasets and additional variables could provide a deeper understanding of performance patterns in cricket.

REFERENCES

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Matplotlib Documentation (2024): **https://matplotlib.org/**

Seaborn Documentation (2024): **https://seaborn.pydata.org/**